

Chapter 6

Momentum

Linear momentum

The momentum $\mathbf{p}$ of a body is defined as:

$$\vec{p} = m\vec{v}$$

$$\vec{p} = p_x \vec{i} + p_y \vec{j} + p_z \vec{k}$$

$$= mv_x \vec{i} + mv_y \vec{j} + mv_z \vec{k}$$

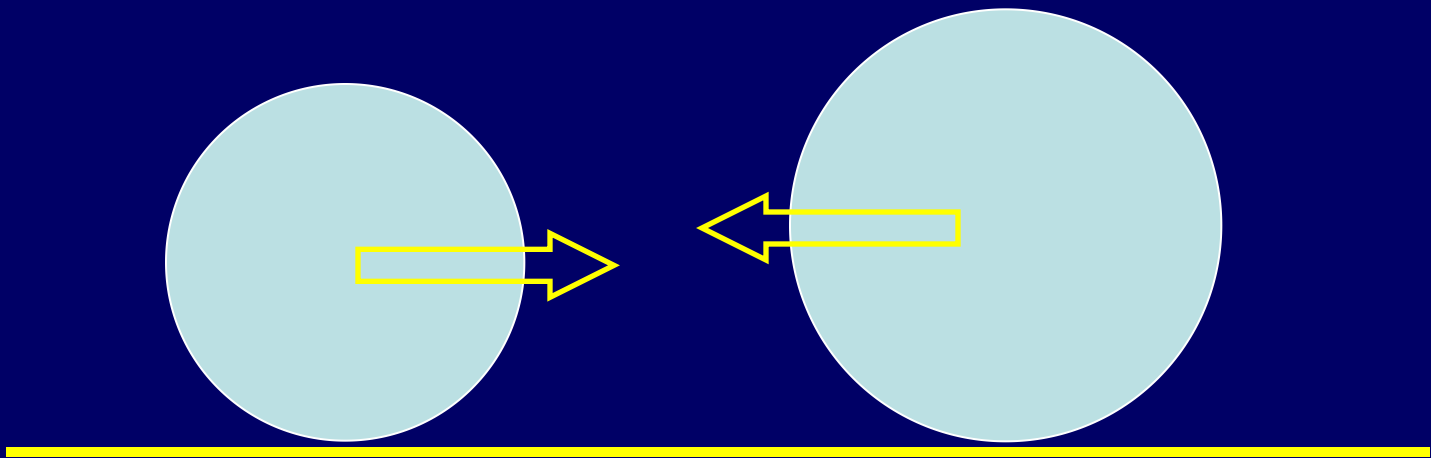
$$\sum \vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt} = \frac{d\vec{p}}{dt}$$

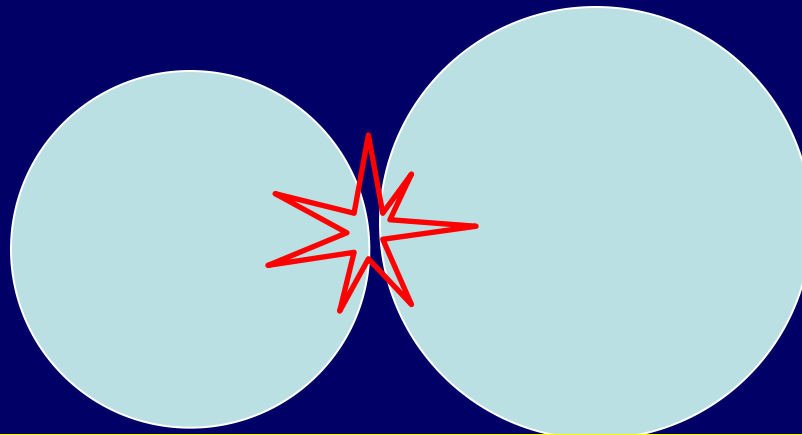
$$\boxed{\sum \vec{F} = \frac{d\vec{p}}{dt}}$$

This is another form of
Newton's second law.

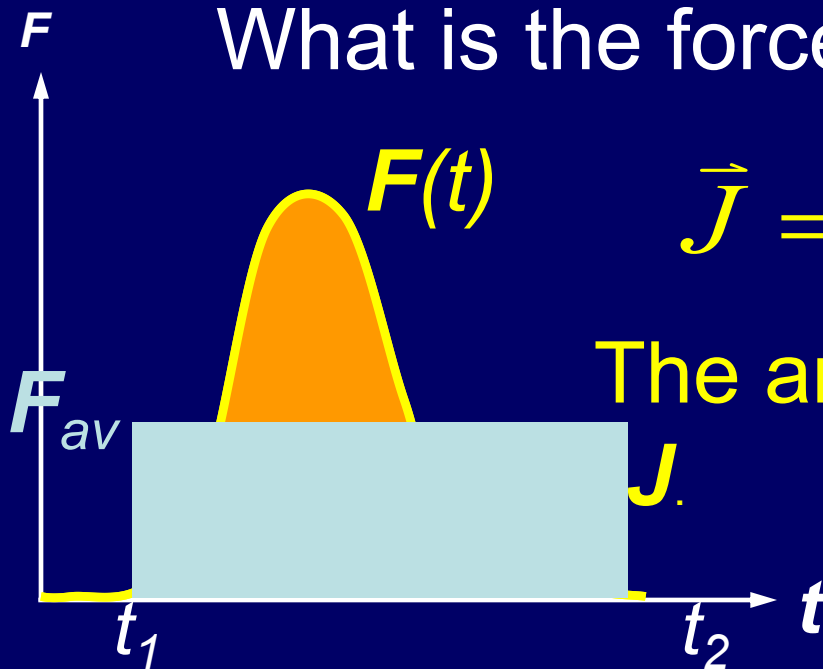
Collision







What is the force between two bodies?



$$\vec{J} = \int_{t_1}^{t_2} \vec{F} dt \quad \text{impulse 冲量}$$

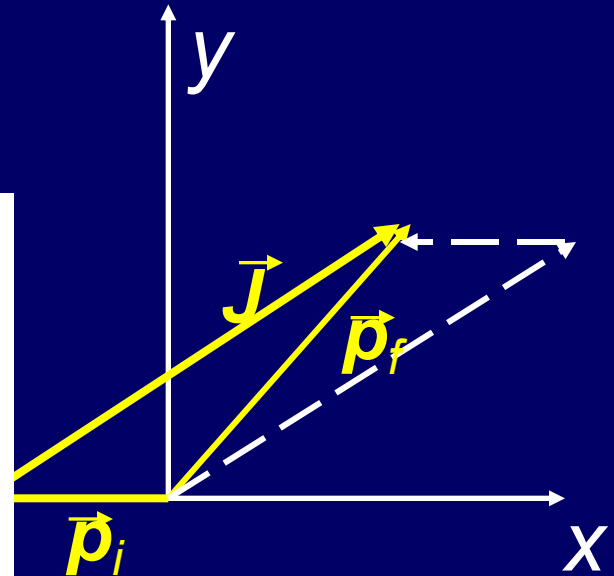
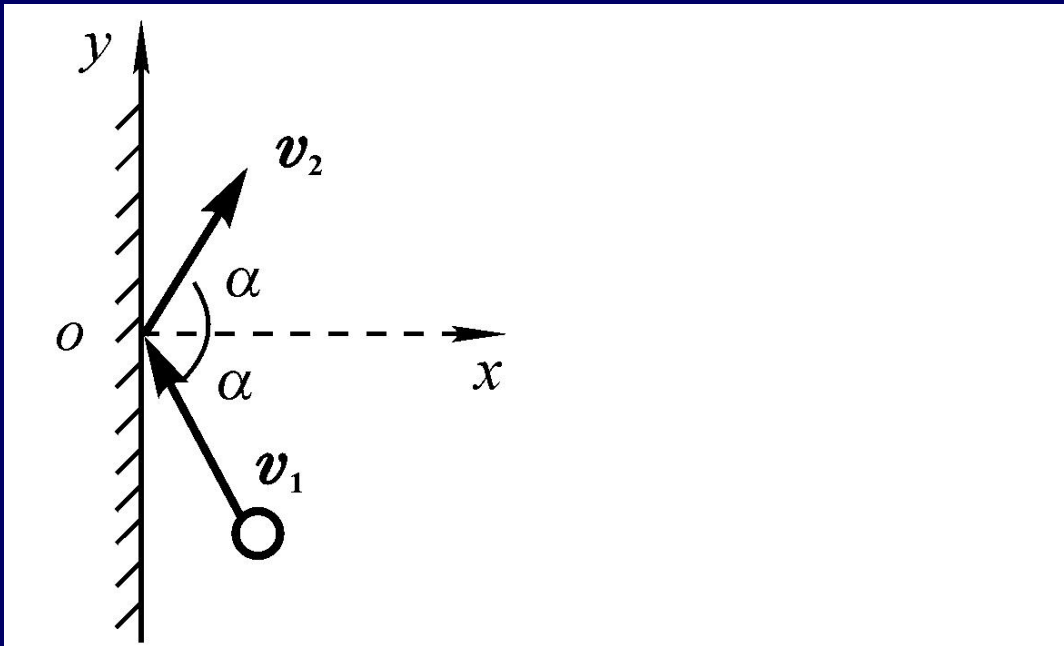
The area under the $F(t)$ curve

$$J = F_{av} (t_2 - t_1) = F_{av} \Delta t$$

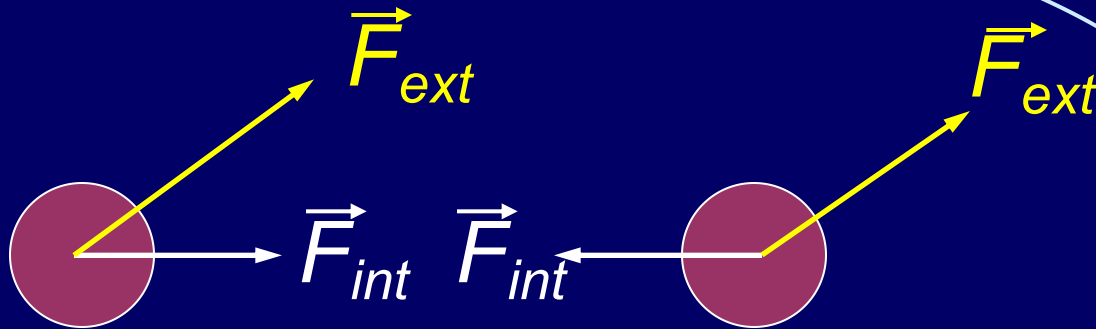
$$\sum \vec{F} = \frac{d\vec{p}}{dt} \quad d\vec{p} = \sum \vec{F} dt$$

$$\vec{p}_f - \vec{p}_i = \int_{p_i}^{p_f} d\vec{p} = \int_{t_1}^{t_2} \sum \vec{F} dt = \vec{J} = F_{av} \Delta t$$

$$\vec{J} = \vec{p}_f - \vec{p}_i = \Delta \vec{p}$$



Conservation of momentum



$$\vec{F}_{ext} = 0$$

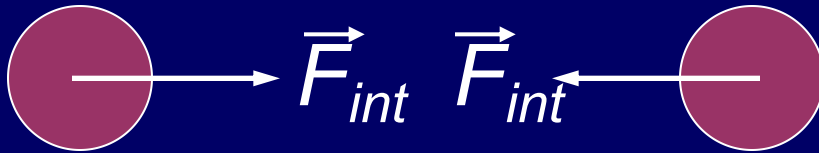
$$\vec{J} = \int_{t_1}^{t_2} \Sigma \vec{F} dt = \int_{t_1}^{t_2} \Sigma (\vec{F}_{int} + \vec{F}_{ext}) dt$$

Conservation of momentum

•The frame of reference.

$$\vec{J} = \Delta \vec{p} = \vec{p}_f - \vec{p}_i = 0$$

•Although p_1 and p_2 may both change, their vector sum stays the same.



$$\vec{F}_{ext} = 0 \quad \vec{p}_f = \vec{p}_i \quad \vec{p}_{1f} + \vec{p}_{2f} = \vec{p}_{1i} + \vec{p}_{2i}$$

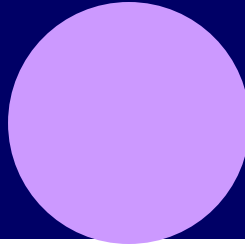
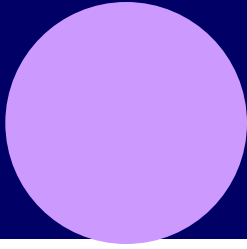
$$\vec{J} = \int_{t_1}^{t_2} \Sigma \vec{F} dt = \int_{t_1}^{t_2} \Sigma (\vec{F}_{int} + \vec{F}_{ext}) dt = \int_{t_1}^{t_2} \Sigma \vec{F}_{int} dt = 0$$

•Conservation of momentum consists of three components formulas.

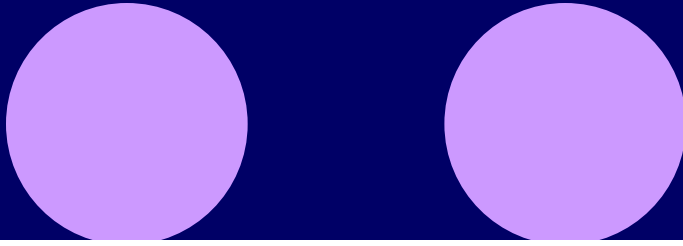
$$\vec{p}_{xf} = \vec{p}_{xi} \quad \vec{p}_{yf} = \vec{p}_{yi}$$

$$\vec{p}_{zf} = \vec{p}_{zi}$$

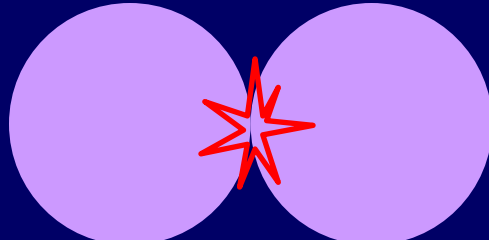
Collision in 1-D



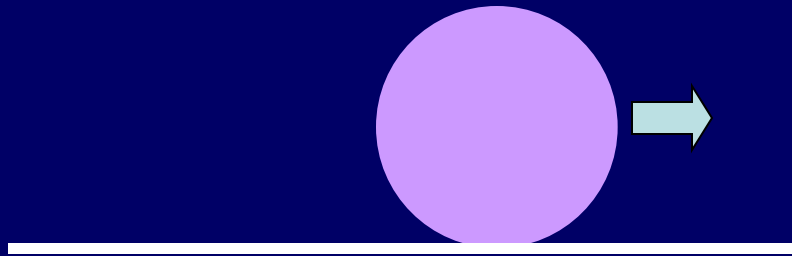
Collision in 1-D



Collision in 1-D



Collision in 1-D

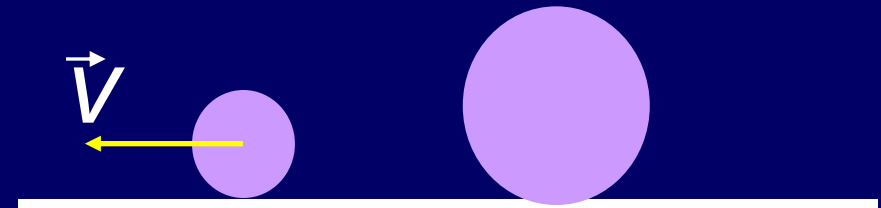
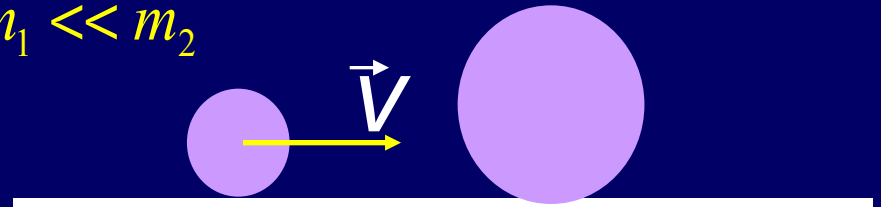


m_1

For elastic collision

m_1

$m_1 \ll m_2$



$$v_{2i} = 0$$

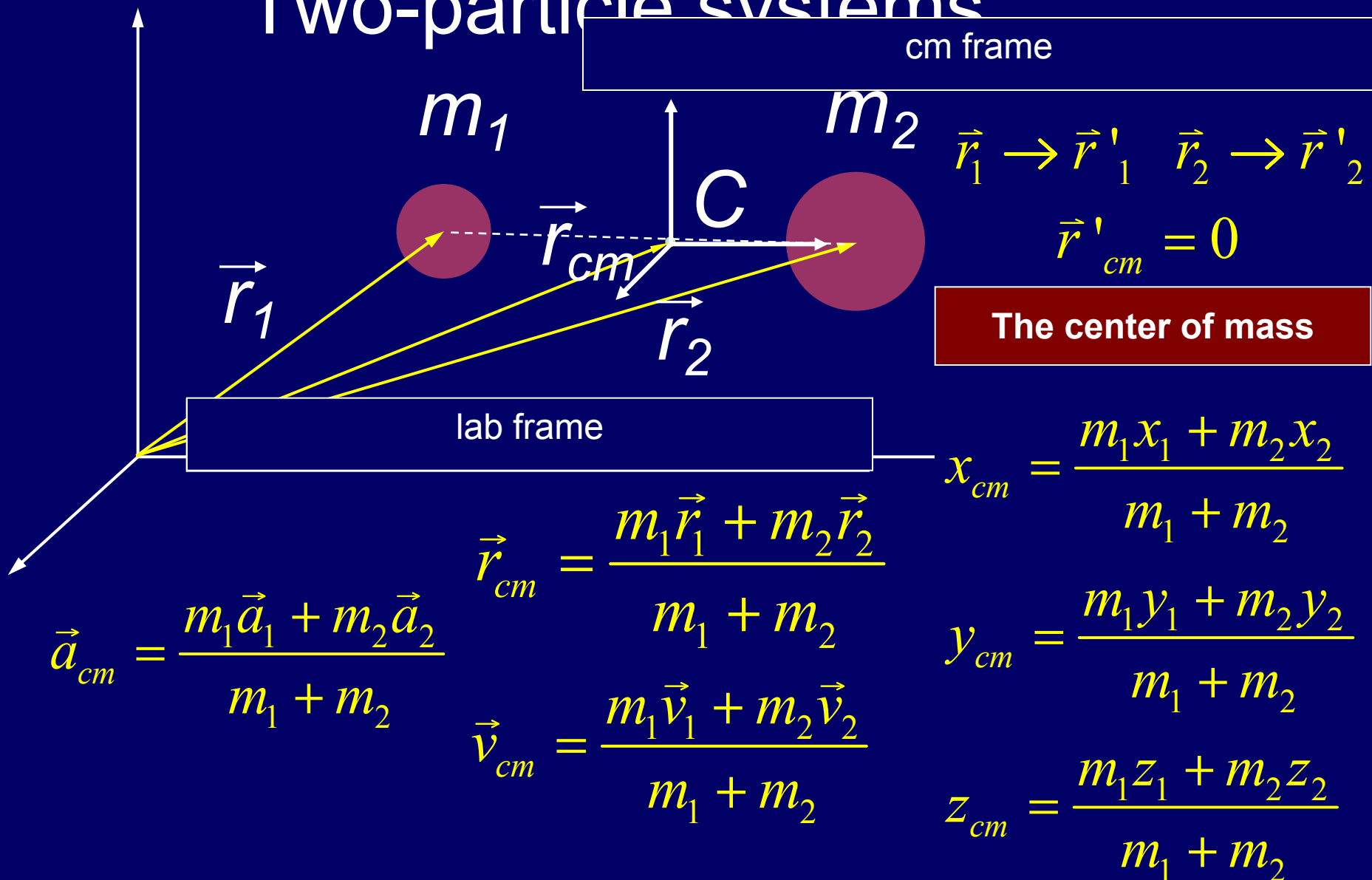
$$v_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) v_{1i} = -v_{1i}$$

$$m_1 = m_2$$

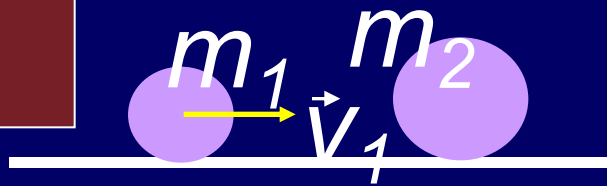
$$m_1 \ll m_2$$

$$v_{2f} = \left(\frac{2m_1}{m_1 + m_2} \right) v_{1i} = 0$$

Two-particle systems

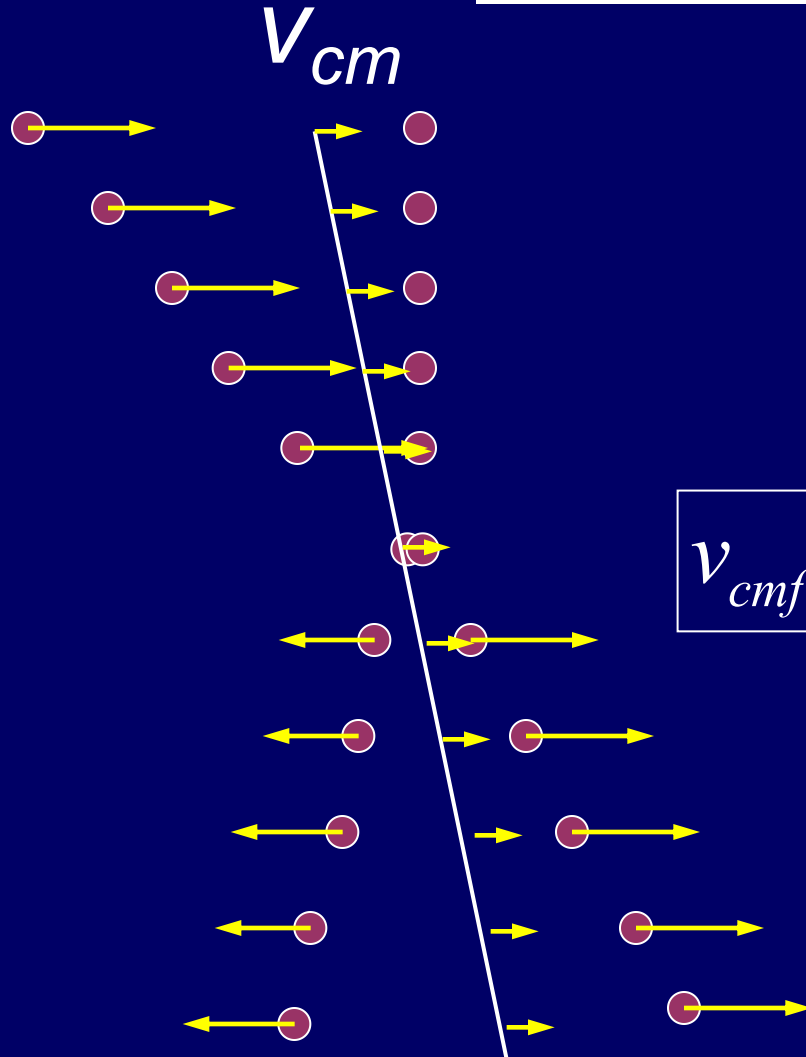


lab frame



$$v_{1i} \neq 0, \quad v_{2i} = 0$$

$$\begin{cases} v_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) v_{1i} \\ v_{2f} = \left(\frac{2m_1}{m_1 + m_2} \right) v_{1i} \end{cases}$$

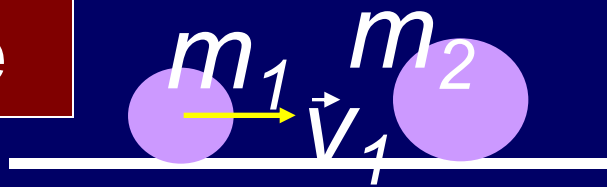


$$v_{cmf} = v_{cm}$$

$$v_{cm} = \frac{m_1 v_{1i}}{m_1 + m_2}$$

$$v_{cmf} = \frac{m_1 v_{1f} + m_2 v_{2f}}{m_1 + m_2}$$

CM Frame



$$v'_{1i} = v_{1i} - v_{cm}$$

$$v'_{2i} = v_{2i} - v_{cm}$$

$$v_{cm} = \frac{m_1 v_{1i}}{m_1 + m_2} \quad v'_{1i} = \frac{m_2 v_{1i}}{m_1 + m_2} \quad v'_{2i} = -\frac{m_1 v_{1i}}{m_1 + m_2}$$

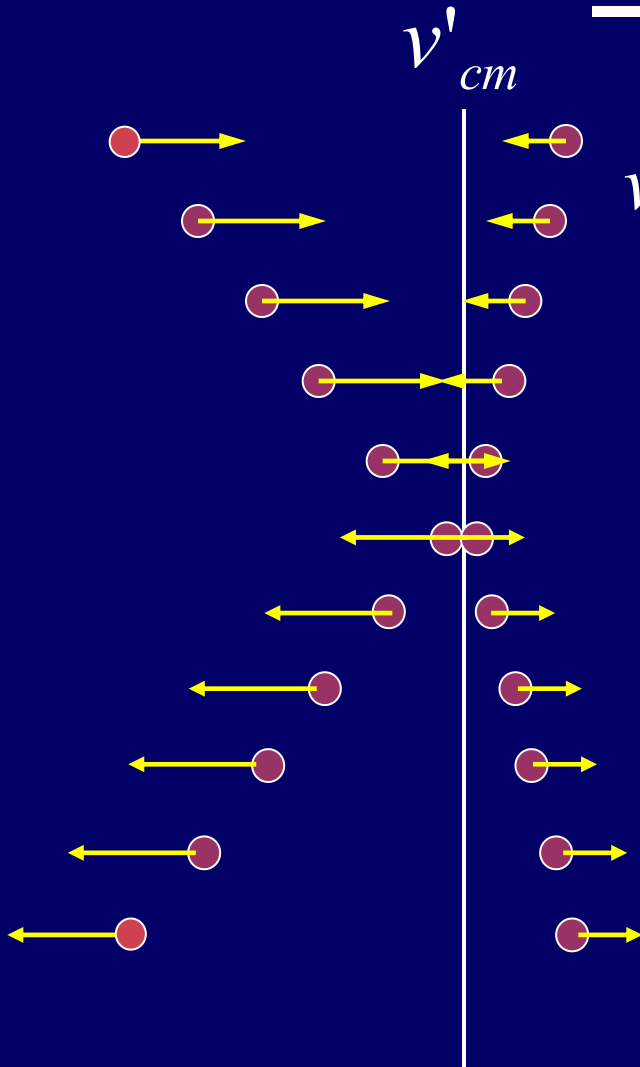
$$v'_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) v'_{1i} + \left(\frac{2m_2}{m_1 + m_2} \right) v'_{2i}$$

$$v'_{2f} = \left(\frac{2m_1}{m_1 + m_2} \right) v'_{1i} + \left(\frac{m_2 - m_1}{m_1 + m_2} \right) v'_{2i}$$

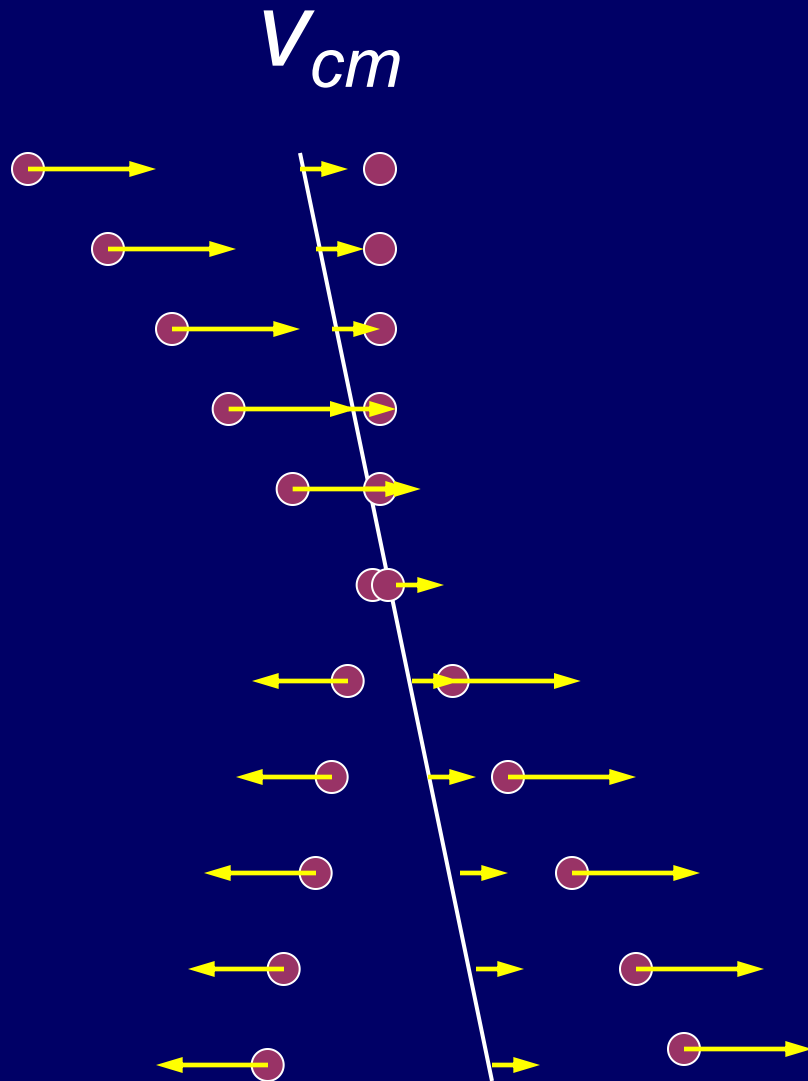
$$v'_{1f} = -\frac{m_2 v_{1i}}{m_1 + m_2} \quad v'_{2f} = \frac{m_1 v_{1i}}{m_1 + m_2}$$

$$v'_{cm} = \frac{m_1 v'_{1i} + m_2 v'_{2i}}{m_1 + m_2} = \frac{m_1 v'_{1f} + m_2 v'_{2f}}{m_1 + m_2}$$

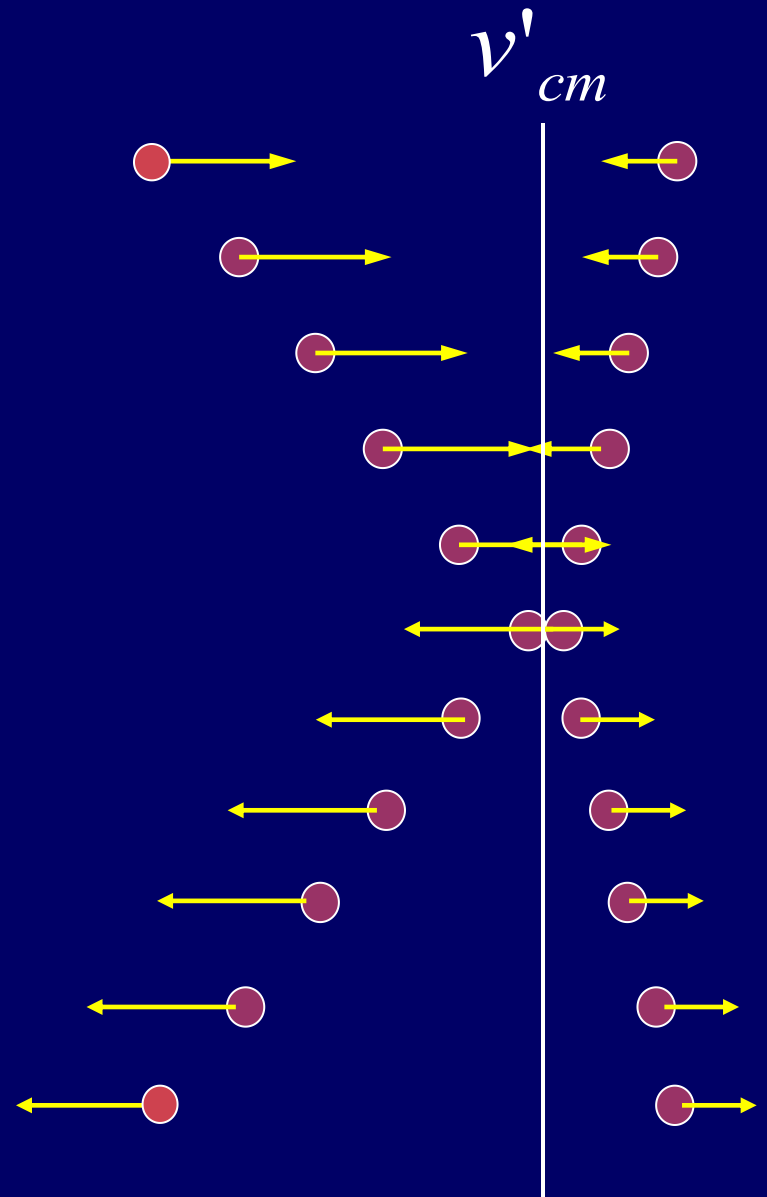
$$= 0$$



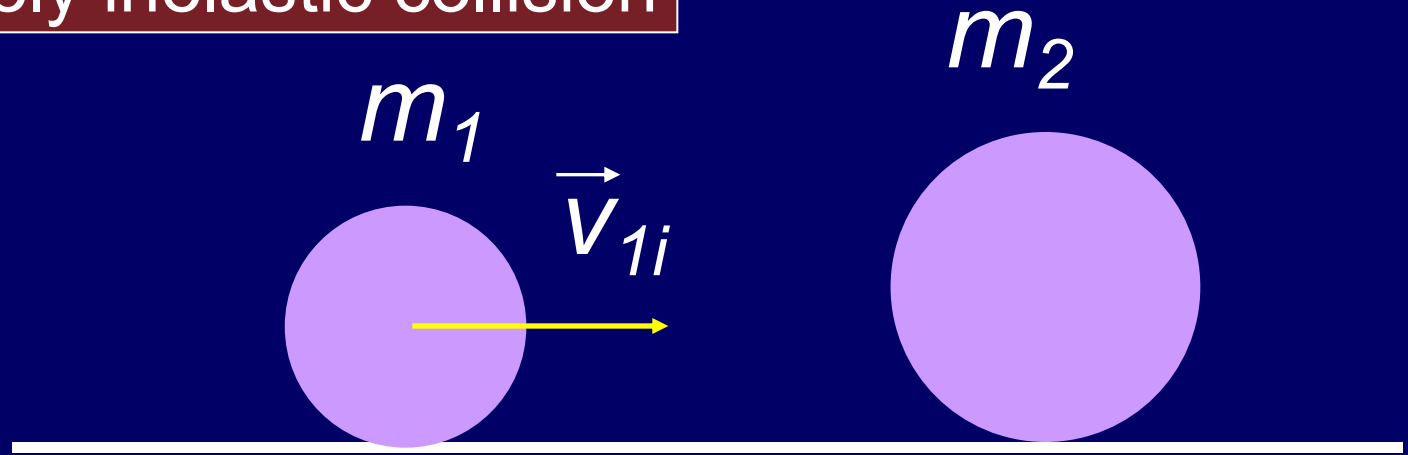
lab frame



cm frame



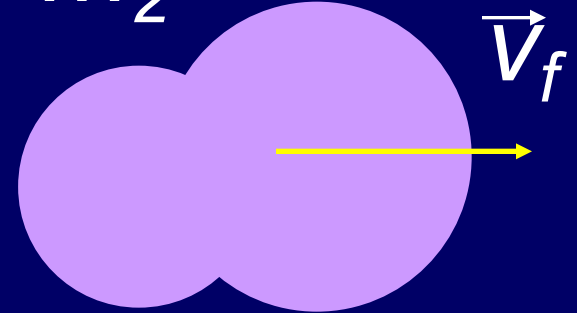
For completely inelastic collision



$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f} \quad m_1 + m_2$$

$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$$

$$v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2} = v_{cm}$$

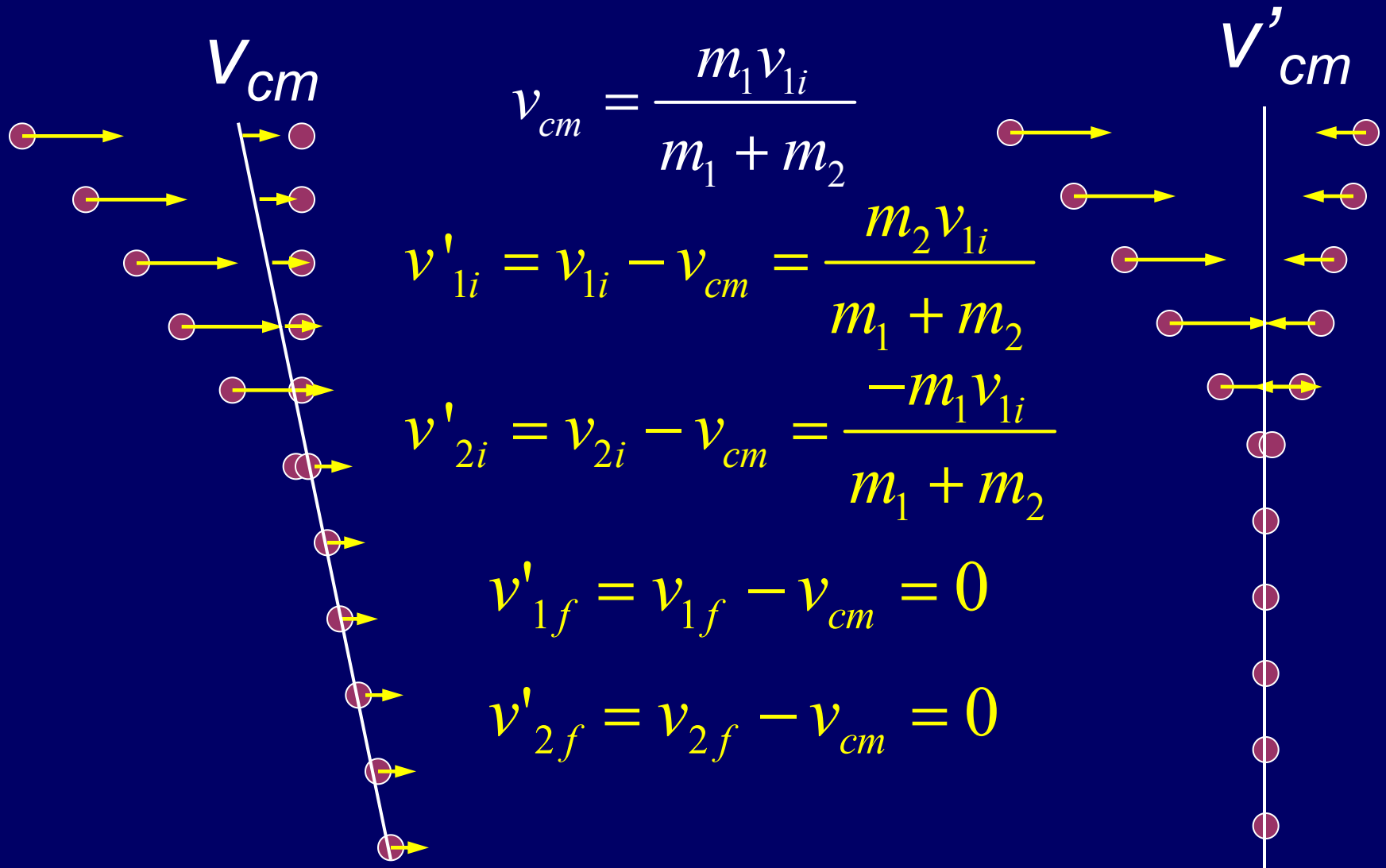


$$v_{2i} = 0$$

$$v_f = \frac{m_1 v_{1i}}{m_1 + m_2} = v_{cm}$$

lab frame

cm frame



Example

in $\vec{i}$ direction

$$p_i = 4 \times 1 = 4$$

$$p_f = p_i = 4$$

$$p_{2f} = m_2 v_{x2f} = 2\sqrt{3}a, \quad p_{1f} = 4 - 2\sqrt{3}a$$

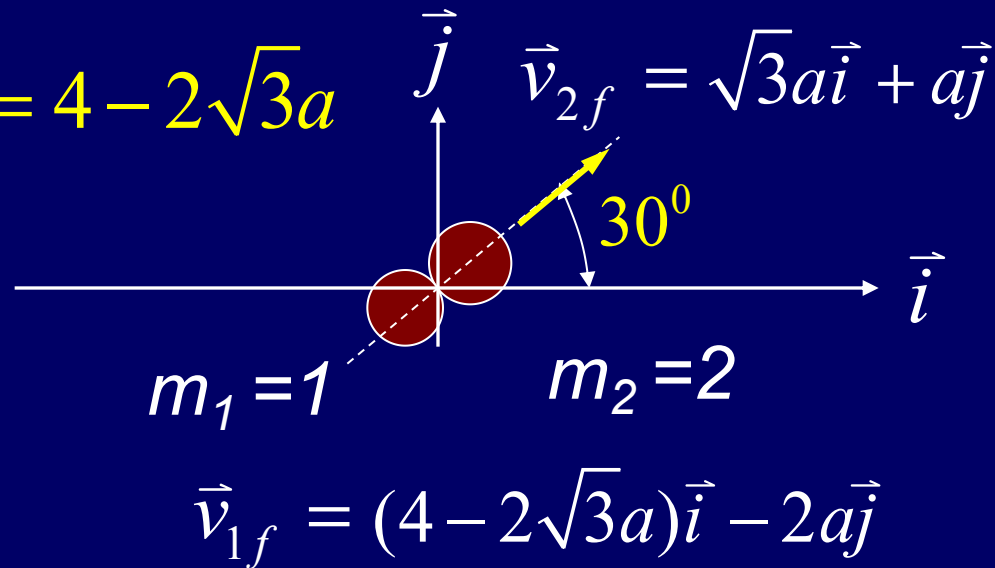
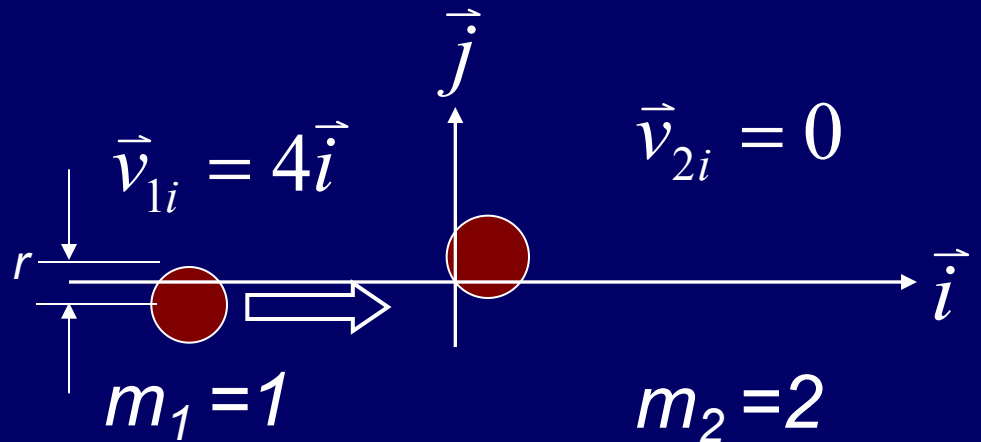
$$v_{1f} = p_{1f} / m_1 = 4 - 2\sqrt{3}a$$

in $\vec{j}$ direction

$$p_i = 0 \quad p_f = p_i = 0$$

$$p_{2f} = 2a, \quad p_{1f} = -2a$$

$$v_{1f} = -2a$$



Example

For elastic collision

$$\frac{1}{2}m_1v_{1i}^2 = \frac{1}{2}m_1v_{1f}^2 + \frac{1}{2}m_2v_{2f}^2$$

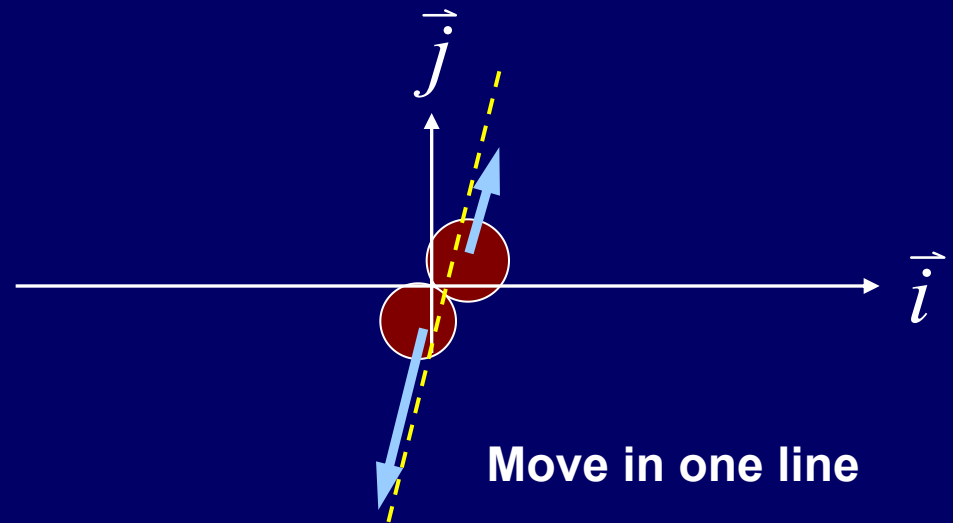
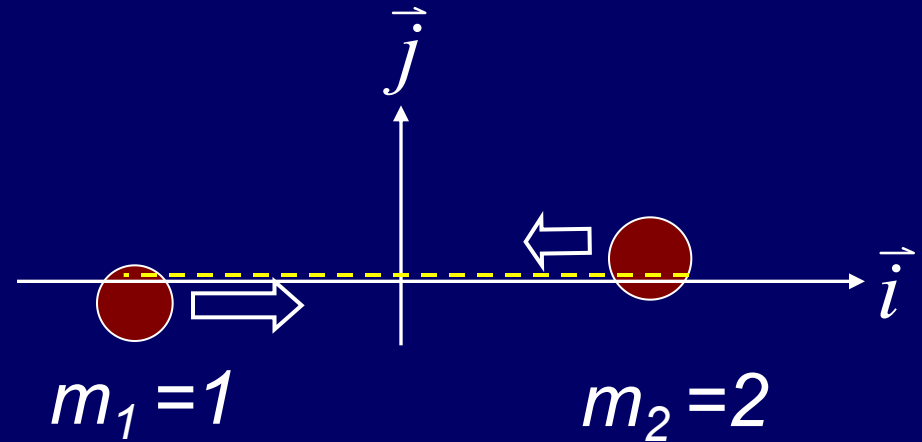
$$8 = 2(2 - \sqrt{3}a)^2 + 2a^2 + 3a^2 +$$

CM Frame

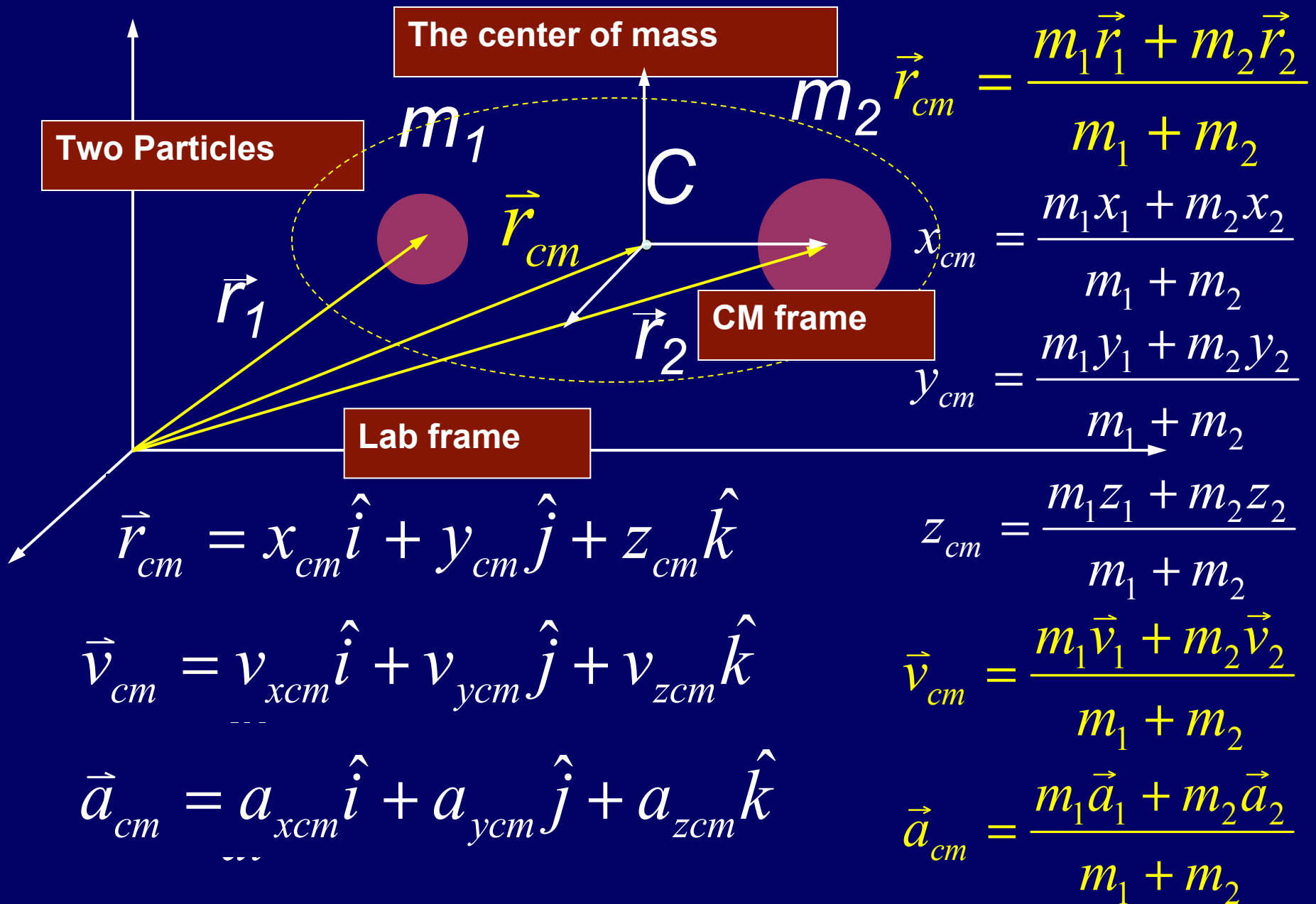
$$\vec{v}_{cm} = \frac{4 \times 1}{1+2} \hat{i} = \frac{4}{3} \hat{i}$$

$$\vec{v}'_{1i} = 2\frac{2}{3}\hat{i} \quad \vec{v}'_{2i} = -\frac{4}{3}\hat{i}$$

$$\vec{v}'_{1f} = -1.33\hat{i} - 2.3\hat{j}$$



$$\vec{v}'_{2f} = 0.67\hat{i} + 1.15\hat{j}$$



Question: A wall moves with $\mathbf{v}$ in the x direction and
A ball moves with $-\mathbf{v}$. What is the velocity of the ball after it
collided the wall. The wall mass is much larger than the ball.

Which one is correct?

(a) $\mathbf{v}$, (b) $2\mathbf{v}$, (c) $3\mathbf{v}$, (d) $4\mathbf{v}$

CHAP. 6

Problems

10, 14, 16, 17,
19 (P136)